20. ROS2 recording and playback tool

1. Introduction to the recording and playback tool

Bag2 is a commonly used recording and playback tool in ROS2, and it is used to record topic data. We can use this command to store topic data as a file. Later on, we do not need to start any node, and we can directly publish the topic data in the bag file.

This tool is extremely useful when we are building a real robot. For example, we can record the topic data of a robot while it is having problems. After the recording is finished, we can publish it many times for testing and experimentation, or we can share the topic data with others to verify algorithms, etc.

We will try to use the bag tool to record topic data and play it back a second time.

2. Usage tutorial

2.1. Starting the topic node to be recorded

For example, the talker in the ros2 demo:

```
ros2 run demo_nodes_py talker
```

2.2. Recording

`/topic-name` is the topic name

```
# Record a single topic
ros2 bag record /topic-name
# Record multiple topics
ros2 bag record topic-name1 topic-name2
# Record all topics
ros2 bag record -a
```

Other options

-o name Customize the output file name

```
ros2 bag record -o file-name topic-name
```

-s Storage format

ROS2 Jazzy uses the MCAP storage format by default, and you can also specify an installed storage plugin with `-s`, for example `sqlite3`:

```
ros2 bag record -s sqlite3 /topic-name
```

2.3. Viewing the information of the recorded topics

Before playing back a video, you can view the relevant information about the video through the file information, such as the time, size, type and number of the topic records

```
# Assume the recorded file is rosbag2_2023_10_31-07_58_23
ros2 bag info rosbag2_2023_10_31-07_58_23
```

2.4. Playing back and viewing

2.4.1. Playback

Next we can play the data back again. The following command can be used to play back the data

```
ros2 bag play rosbag2_2023_10_31-07_58_23
```

2.4.2. Viewing

Use the topic command of ros2 to view the data

```
ros2 topic echo /chatter
```

2.4.3. Playback options

1. Speed playback -r

The -r option can modify the playback rate. For example, the -r value, for example -r 10, means 10 times the speed, playing the topic ten times faster

```
ros2 bag play rosbag2_2023_10_31-07_58_23 -r 10
```

2. Loop playback -l

This is the one-song loop

```
ros2 bag play rosbag2_2023_10_31-07_58_23 -l
```

3. Playing back a single topic

```
ros2 bag play rosbag2_2023_10_31-07_58_23 --topics /chatter
```

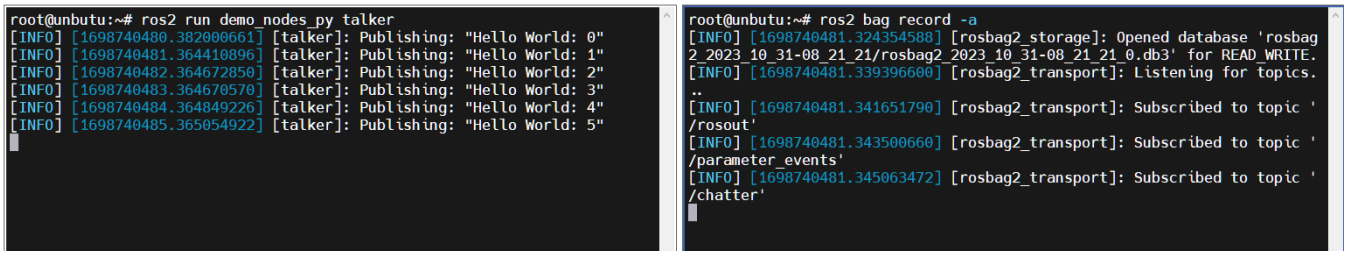
3. Example

3.1. Running the talker node

```
ros2 run demo_nodes_py talker
```

3.2. Recording

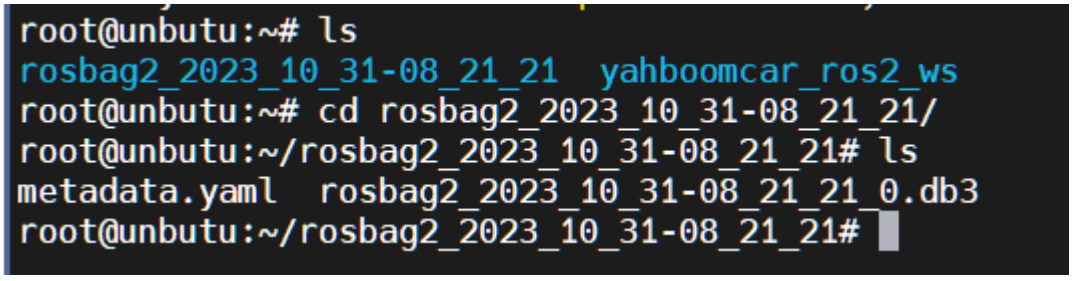
```
# Record all topics
ros2 bag record -a
```



How do we stop the recording? We simply press `Ctrl+C` in the terminal to interrupt the recording.

Then you will find a folder in the terminal, named `rosbag2_2023_10_31-08_21_21`

Open the folder and you can see its contents



This completes the recording.

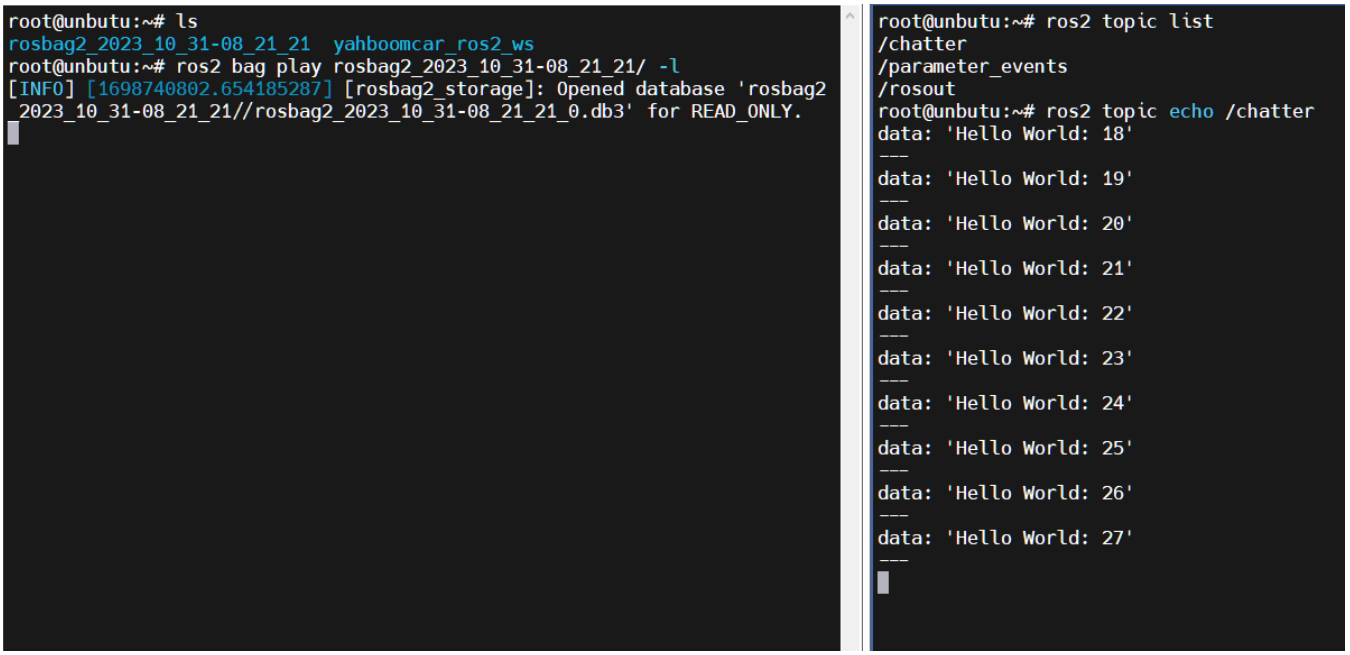
3.3. Playing back and viewing

Here we play it back in a loop

```
ros2 bag play rosbag2_2023_10_31-07_58_23 -l
```

Open another terminal to view the topic:

```
ros2 topic echo /chatter
```


root@unbutu:~# ls
rosbag2_2023_10_31-08_21_21 yahboomcar_ros2_ws
root@unbutu:~# ros2 bag play rosbag2_2023_10_31-08_21_21/ -l
[INFO] [1698740802.654185287] [rosbag2_storage]: Opened database 'rosbag2_2023_10_31-08_21_21//rosbag2_2023_10_31-08_21_21_0.db3' for READ_ONLY.
root@unbutu:~# ros2 topic list
/chatter
/parameter_events
/rosout
root@unbutu:~# ros2 topic echo /chatter
data: 'Hello World: 18'
---
data: 'Hello World: 19'
---
data: 'Hello World: 20'
---
data: 'Hello World: 21'
---
data: 'Hello World: 22'
---
data: 'Hello World: 23'
---
data: 'Hello World: 24'
---
data: 'Hello World: 25'
---
data: 'Hello World: 26'
---
data: 'Hello World: 27'
